

UNIT-III:-

INNER PRODUCT SPACES:-

Introduction:-

In the previous units, the defn & Properties of vector spaces were dealt with. But geometrical notions such as length of vector, angle between two vectors were not discussed. This unit focuses on these aspects and hence only real (or) complex vector spaces will be dealt with.

In this unit F will be either
Real Field $\mathbb{R}$ (or) Complex Field $\mathbb{C}$.

Before going ahead, here is a recall of Properties of
Complex numbers

- (1) If $z \in \mathbb{C}$, then $z = x + iy$, where $x, y \in \mathbb{R}$,
 x is Real part of z , & y is Imaginary part of z .

$$(i) \quad x = \operatorname{Re}(z), \quad y = \operatorname{Im}(z)$$

(2) $|z| = \sqrt{x^2 + (y)^2}$

$$\& \quad \bar{z} = \text{Conjugate of } z \\ = x - iy$$

$$\& \quad |z| = |\bar{z}|$$

(3) $z + \bar{z} = 2x = 2\operatorname{Re}(z)$

(4) $z - \bar{z} = 2yi = 2i\operatorname{Im}(z)$

$$(5) \quad z \cdot \bar{z} = |z|^2 = x^2 + y^2$$

$$(6) \quad \overline{\bar{z}} = z$$

$$7) \quad |z| = \sqrt{x^2 + y^2} \geq x$$

$$\Rightarrow \boxed{|z| \geq \operatorname{Re}(z)}$$

8) If z_1, z_2 are complex numbers, then

$$(i) \quad |z_1 + z_2| \leq |z_1| + |z_2| \quad (\text{Triangle inequality})$$

$$(ii) \quad \overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$$

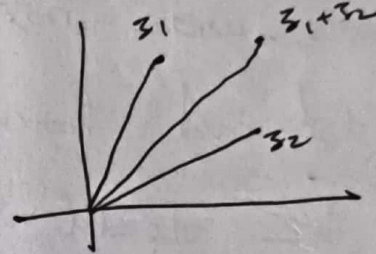
$$(iii) \quad \overline{z_1 z_2} = (\bar{z}_1)(\bar{z}_2)$$

$$(iv) \quad \overline{\left(\frac{z_1}{z_2}\right)} = \bar{z}_1 / \bar{z}_2 \quad \text{if } \bar{z}_2 \neq 0.$$

$$(v) \quad |z_1 z_2| = |z_1| |z_2|$$

$$(vi) \quad \left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|} \quad \text{if } z_2 \neq 0$$

$$(vii) \quad |z_1 - z_2| \geq ||z_1| - |z_2||$$



Defn: Inner Product

Let V be a vector space over F . An inner product on V is a function that assigns to every ordered pair of vectors x & y in V , a scalar in F , denoted by $\langle x, y \rangle$, such that for all x, y & $z \in V$ and $\forall c \in F$, the following hold:

- (a) $\langle x+y, z \rangle = \langle x, z \rangle + \langle y, z \rangle$
 - (b) $\langle cx, y \rangle = c \langle x, y \rangle$
 - (c) $\overline{\langle x, y \rangle} = \langle y, x \rangle$, where bar denotes, the complex conjugate.
 - (d) $\langle x, x \rangle \geq 0$ if $x \neq 0$ [Non-negativity Property]
- These two axioms show Inner Product is linear in the first component

Defn: Inner Product Space

A vector space V over F is said to be an inner product space w.r.t the specified inner product defined on it.

Note:-

1) If $a_1, a_2, \dots, a_n \in F$ and $y, v_1, v_2, \dots, v_n \in V$, Then

$$\left\langle \sum_{i=1}^n a_i v_i, y \right\rangle = \sum_{i=1}^n a_i \langle v_i, y \rangle$$

Proof:-

$$\left\langle \sum_{i=1}^n a_i v_i, y \right\rangle = \left\langle a_1 v_1 + a_2 v_2 + \dots + a_n v_n, y \right\rangle$$

$$= \langle a_1 v_1, y \rangle + \langle a_2 v_2, y \rangle + \dots + \langle a_n v_n, y \rangle$$

$$= a_1 \langle v_1, y \rangle + a_2 \langle v_2, y \rangle + \dots + a_n \langle v_n, y \rangle$$

$$= \sum_{i=1}^n a_i \langle v_i, y \rangle$$

Note 2:-

Let $V(F)$ be a Inner Product Space, then $\langle 0, 0 \rangle = 0$

Proof:-

As $0 \in V$, $0 \in F$,

By scalar multiplication, $0 \cdot 0 = 0 \in V$

$$\text{Now, } \langle 0, 0 \rangle = \langle 0 \cdot 0, 0 \rangle$$

$$= 0 \langle 0, 0 \rangle$$

$$= 0, \text{ Since } \langle 0, 0 \rangle \text{ be a scalar in } F$$

$$\& 0 \in F$$

— x —

Euclidean Space:-

If F is the field of real numbers, then ~~$V(F)$~~ is the Inner Product space $V(F)$ is called Real IPS
(or) Euclidean space. Here $\langle x, y \rangle$ is a real number.

Unitary space:-

If F is the field of complex numbers, then a Inner Product space $V(F)$ is called Complex IPS
(or) Unitary space. Here $\langle x, y \rangle$ is a complex number.

Problem 1. [Standard Inner Product]

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Let $x = (a_1, a_2, \dots, a_n) \in y = (b_1, b_2, \dots, b_n)$ be in F^n

Then define

$$\langle x, y \rangle = \sum_{i=1}^n a_i \overline{b_i} \quad \text{--- (1)}$$

Verify that $\langle \cdot, \cdot \rangle$ is an inner product.

Proof/Solution..

S.T $F^n(C)$ is an inner product space w.r.t

T.P. defined as

$$\langle x, y \rangle = \sum_{i=1}^n a_i \overline{b_i}$$

where $x = (a_1, \dots, a_n) \in F^n$
 $y = (b_1, \dots, b_n) \in F^n$

(a) Let $z = (c_1, c_2, \dots, c_n) \in F^n$,

$$x + y = (a_1 + b_1, a_2 + b_2, \dots, a_n + b_n)$$

$$a) \langle x + y, z \rangle = \sum_{i=1}^n (a_i + b_i) \overline{c_i}$$

$$= \sum_{i=1}^n [a_i \overline{c_i} + b_i \overline{c_i}]$$

$$= \sum_{i=1}^n a_i \overline{c_i} + \sum_{i=1}^n b_i \overline{c_i}$$

$$= \langle x, z \rangle + \langle y, z \rangle$$

$$(b) \langle cx, y \rangle = \sum_{i=1}^n (ca_i) \overline{b_i}$$

$$= c \sum_{i=1}^n a_i \overline{b_i}$$

$$= c \langle x, y \rangle$$

(c) From (1)

$$\langle y, x \rangle = \sum_{i=1}^n b_i \overline{a_i} \quad \text{--- (2)}$$

$$\text{Now } \overline{\langle x, y \rangle} = \overline{\left(\sum_{i=1}^n a_i \overline{b_i} \right)}$$

$$\begin{aligned}
&= \sum_{i=1}^n \overline{a_i b_i} \\
&= (\overline{a_1 b_1} + \overline{a_2 b_2} + \dots + \overline{a_n b_n}) \quad \overline{\delta_1 + \delta_2} = \overline{\delta_1} + \overline{\delta_2} \\
&= \overline{a_1 b_1} + \overline{a_2 b_2} + \dots + \overline{a_n b_n} \quad \overline{\delta_1 \delta_2} = \overline{\delta_1} \overline{\delta_2} \\
&= \overline{a_1} \overline{b_1} + \overline{a_2} \overline{b_2} + \dots + \overline{a_n} \overline{b_n} \quad \overline{\delta_1} = \delta_1 \\
&= b_1 \overline{a_1} + b_2 \overline{a_2} + \dots + b_n \overline{a_n} \quad (\text{Commutative property}) \\
&= \sum_{i=1}^n b_i \overline{a_i}
\end{aligned}$$

$$= \langle y, x \rangle \quad (\text{by } \textcircled{2})$$

(d) If $x \neq 0$, $x = (a_1, a_2, \dots, a_n)$ (at least one $a_i \neq 0$)

$$\begin{aligned}
\langle x, x \rangle &= \sum_{i=1}^n a_i \overline{a_i} \\
&= \sum_{i=1}^n |a_i|^2, \text{ since } |a_i| \geq 0 \text{ for some } i \text{ (not all zero)} \\
&> 0 \quad \left(\begin{array}{l} \text{Sum of } n \text{ non-negative real numbers} \\ \text{and here is } +ve. \end{array} \right)
\end{aligned}$$

$\overline{\delta \cdot \overline{\delta}} = |\delta|^2$

Thus $\langle x, x \rangle > 0$ if $x \neq 0$

Hence the Product defined $\textcircled{1}$ is an inner Product
On F^n w.r.t the inner Product, F^n is an inner Product
Space.

Problem 2: If $\langle x, y \rangle$ is any inner Product on a
vector space V and $\delta > 0$, we define another inner Product
by the rule $\langle x, y \rangle' = \delta \langle x, y \rangle$

Verify that $\langle \cdot, \cdot \rangle'$ is an inner Product.

Problem 3:-

Let $V = C([0,1])$ denote the vector space of real valued continuous ~~complex~~ functions on $[0,1]$, for $f, g \in V$

Let $\langle f, g \rangle = \int_0^1 f(x) \overline{g(x)} dx$. — (1) p.t. that $\langle f, g \rangle = \int_0^1 f(x) \overline{g(x)} dx$ is an I.P. where $f, g \in C([0,1])$ & V is a v.s. of complex valued continuous functions $[0,1]$

Verify that $\langle \cdot, \cdot \rangle$ is an inner product

Soln:- Let $h \in V$

(a) $\langle f+h, g \rangle = \int_0^1 [f(x) + h(x)] \overline{g(x)} dx$

$$= \int_0^1 [f(x) \overline{g(x)} + h(x) \overline{g(x)}] dx$$

$$= \int_0^1 f(x) \overline{g(x)} dx + \int_0^1 h(x) \overline{g(x)} dx$$

$$= \langle f, g \rangle + \langle h, g \rangle$$

Diagram: A number line from 0 to 1. A point is marked at 0.5. A function is defined as $f(x) = x + i x^2 = g(x) + i h(x)$.

(b) $\langle cf, g \rangle = \int_0^1 cf(x) \overline{g(x)} dx$

$$= c \int_0^1 f(x) \overline{g(x)} dx$$

$$= c \langle f, g \rangle$$

(c) $\overline{\langle f, g \rangle} = \overline{\int_0^1 f(x) \overline{g(x)} dx}$

$$= \int_0^1 \overline{f(x) \overline{g(x)}} dx$$

$$= \int_0^1 \overline{f(x)} g(x) dx$$

Diagram: A number line from 0 to 1. A point is marked at 0.5. A function is defined as $f(x) = x + i x^2 = g(x) + i h(x)$.

$$= \int_0^1 g(x) \overline{f(x)} dx$$

$$= \langle g, f \rangle$$

For $f \neq 0$,
 $\downarrow$ $\langle f, f \rangle = \int_0^1 f(x) \overline{f(x)} dx$

$$= \int_0^1 |f(x)|^2 dx, \text{ Since } |f(x)| \geq 0$$

$$> 0$$

Thus ① is an inner Product on $V = C[0,1]$

Hence, $V = C[0,1]$ is a inner Product Space

— x —

Defn. Conjugate Transpose (or) Adjoint of a matrix.

Let $A \in M_{m \times n}(F)$. We define the Conjugate transpose (or) adjoint of A , denoted by A^* is defined

as $A^* = \overline{A^T}$

Example:-

Let $A = \begin{bmatrix} -i & 2 \\ 3-4i & 1-2i \end{bmatrix}$ Then $A^* = \begin{bmatrix} i & 3+4i \\ 2 & 1+2i \end{bmatrix}$

Hence (1) is an inner product on $M_{n \times n}(F)$

Hence $M_{n \times n}(F)$ is an inner product space & is called
Frobenius inner product.

Properties of inner product:

Let $V(F)$ be an inner product space. Then for
 $x, y, z \in V$, $\alpha \in F$, the following statements are true.

$$(a) \langle x, y+z \rangle = \langle x, y \rangle + \langle x, z \rangle$$

$$(b) \langle x, \alpha y \rangle = \overline{\alpha} \langle x, y \rangle$$

$$(c) \langle x, 0 \rangle = \langle 0, x \rangle = 0$$

$$(d) \langle x, x \rangle = 0 \text{ iff } x = 0$$

$$(e) \langle x, y \rangle = \langle x, z \rangle \text{ for all } x \in V, \text{ then } y = z.$$

Proof:

$$\begin{aligned} (a) \langle x, y+z \rangle &= \overline{\langle y+z, x \rangle} \\ &= \overline{\langle y, x \rangle + \langle z, x \rangle} \\ &= \overline{\langle y, x \rangle} + \overline{\langle z, x \rangle} \\ &= \langle x, y \rangle + \langle x, z \rangle \end{aligned}$$

$$\begin{aligned} (b) \langle x, \alpha y \rangle &= \overline{\langle \alpha y, x \rangle} \\ &= \overline{\alpha \langle y, x \rangle} \\ &= \overline{\alpha} \overline{\langle y, x \rangle} \\ &= \overline{\alpha} \langle x, y \rangle \end{aligned}$$

$$\begin{aligned}
 (c) \quad \langle x, 0 \rangle &= \langle x, a+(-a) \rangle, \quad a \in V \\
 &= \langle x, a \rangle + \langle x, -a \rangle \\
 &= \langle x, a \rangle - \langle x, a \rangle \\
 &= 0
 \end{aligned}$$

$$\text{By } \langle 0, x \rangle = 0$$

(d) Assume that $\langle x, x \rangle = 0 \implies$

$\implies x=0$, if not (ie) $x \neq 0 \implies \langle x, x \rangle > 0$ (4th Axiom of IPS)

Converse Part:-

Assume $x=0$,

Then by (c)

$$\langle x, x \rangle = \langle 0, 0 \rangle = 0$$

(e) Let $\langle x, y \rangle = \langle x, z \rangle$

$$\implies \langle x, y \rangle - \langle x, z \rangle = 0$$

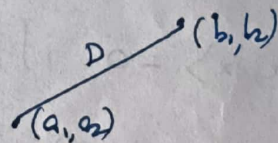
$$\implies \langle x, y \rangle + \langle x, -z \rangle = 0$$

$$\implies \langle x, y-z \rangle = 0 \quad \forall x \in V$$

Then by (c)

$$y-z=0$$

$$\implies \boxed{y=z}$$



$$D = \sqrt{(a_1 - b_1)^2 + (a_2 - b_2)^2}$$

Distance of a_1, a_2 from origin $(0,0)$

$$D = \sqrt{a_1^2 + a_2^2}$$

Defn:- Norm (or) Length of a vector in an Inner Product Space:-

Let V be an inner Product space. For $x \in V$, we define the norm (or) length of x , written as $\|x\|$ is defined as the positive square root of $\langle x, x \rangle$ (ie) $\|x\| = \sqrt{\langle x, x \rangle}$
 $\|x\|^2 = \langle x, x \rangle$

Example:.

Let $V = F^n$ if $x = (a_1, a_2, \dots, a_n)$, then

$$\|x\| = \sqrt{\langle x, x \rangle} = \left[\sum_{i=1}^n a_i \overline{a_i} \right]^{\frac{1}{2}} \quad (\text{by std Inner Product})$$

$$= \left[\sum_{i=1}^n |a_i|^2 \right]^{\frac{1}{2}}$$

$$= \sqrt{|a_1|^2 + |a_2|^2 + \dots + |a_n|^2}, \text{ is the}$$

length of vector $x = (a_1, a_2, \dots, a_n)$ in the n -dimensional Euclidean space.

Property of Norm:.

Let V be an inner product space over F . Then for all $x, y \in V$ and $c \in F$, the following statements are true

(a) $\|cx\| = |c| \|x\|$

(b) $\|x\| = 0$ iff $x = 0$, In (or) $\|x\| \geq 0$

Proof:

(a) $\|cx\|^2 = \langle cx, cx \rangle$

$$= c \overline{c} \langle x, x \rangle$$

$$= |c|^2 \|x\|^2$$

$$= (|c| \|x\|)^2$$

$$\Rightarrow \boxed{\|cx\| = |c| \|x\|}$$

(b) Assume $\|x\| = 0 \Leftrightarrow \sqrt{\langle x, x \rangle} = 0$

$\Leftrightarrow \langle x, x \rangle = 0$

$\Leftrightarrow \boxed{x=0}$ (By Property of IPS.)

Converse part,

~~Assume $x=0 \Rightarrow \langle x, x \rangle = 0$~~

In any case,

$\|x\| = \sqrt{\langle x, x \rangle}$

$\Rightarrow \|x\|^2 = \langle x, x \rangle$

$\Rightarrow \|x\|^2 \geq 0$ ($\because \langle x, x \rangle \geq 0$ if $x \neq 0$)

$\Rightarrow \|x\| \geq 0$

————— x —————

Defn.: Unit vector:-

Let V be an inner Product Space.

If $x \in V$ is such that $\|x\| = 1$, then x is called a unit vector.

Thus, in an inner product space, a vector is called a unit vector if its length is 1.

————— x —————

Result:-

If x is any non-zero vector of an inner product space V , then $\frac{1}{\|x\|}x$ is a unit vector in V .

Proof:

Proof:

As $x \neq 0$, $\|x\| \neq 0$, $\frac{1}{\|x\|}x$ is a real scalar (length)

By scalar multiplication,

$$\frac{1}{\|x\|}x \in V$$

Now

$$\left\langle \frac{1}{\|x\|}x, \frac{1}{\|x\|}x \right\rangle = \frac{1}{\|x\|} \left\langle x, \frac{1}{\|x\|}x \right\rangle \quad (\because \text{Axiom (a)})$$

$$= \frac{1}{\|x\|^2} \langle x, x \rangle \quad (\because \text{by (2) Property of Inner Product})$$

$$= \frac{1}{\|x\|^2} \|x\|^2$$

$$= 1$$

$\Rightarrow \frac{1}{\|x\|}x$ is a unit vector in V

Example:

If $x = (2, 1, 2)$ is a vector in $V_3(\mathbb{R})$ with standard inner product, then

$$(a) \|x\| = \sqrt{\langle x, x \rangle} = \sqrt{a_1^2 + a_2^2 + a_3^2} = \sqrt{4 + 1 + 4} = \sqrt{9} = 3$$

$$(b) \frac{1}{\|x\|}x = \frac{1}{3}(2, 1, 2) = \left(\frac{2}{3}, \frac{1}{3}, \frac{2}{3}\right) \text{ is a unit vector}$$

$$\begin{aligned} \text{Since } \left\| \frac{1}{\|x\|}x \right\| &= \sqrt{\left(\frac{2}{3}\right)^2 + \left(\frac{1}{3}\right)^2 + \left(\frac{2}{3}\right)^2} = \sqrt{\frac{4}{9} + \frac{1}{9} + \frac{4}{9}} \\ &= \sqrt{\frac{9}{9}} \\ &= 1 \end{aligned}$$

Some Important Inequalities & Equality: Identity

①

Cauchy-Schwarz Inequality

$$\left[\begin{array}{l} \text{Pre require } \|x\| > 0 \\ 1. \|x\| > 0 \\ 2. \langle 0, x \rangle = 0 \\ 3. \langle x, y \rangle = \text{is a scalar} \end{array} \right] \left\{ \begin{array}{l} 4. \bar{z} \cdot \bar{z} = |z|^2 \\ 5. |z| = |\bar{z}| \end{array} \right.$$

In an inner product space $V(F)$, then

$$|\langle x, y \rangle| \leq \|x\| \|y\| \quad \text{for } x, y \in V$$

Proof:

Case (i):

Suppose $x = 0$, then $\|x\| = 0$

$$\therefore \|x\| \|y\| = 0 \quad \text{--- (1)}$$

Now Consider (LHS)

$$|\langle x, y \rangle| = |\langle 0, y \rangle|$$

$$= |0| \quad (\because \text{by Property of inner product})$$

$$= 0 \quad \text{--- (2)}$$

From (1) & (2), we have

$$|\langle x, y \rangle| = \|x\| \|y\|.$$

Case (ii):

Suppose $x \neq 0$, then $\|x\| > 0$

Thus $\frac{1}{\|x\|^2}$ is a +ve real number.

Now Define a vector,

$$z = y - \frac{\langle y, x \rangle}{\|x\|^2} x \quad \text{in } V$$

Consider,

$$\|z\|^2 = \langle z, z \rangle = \left\langle y - \frac{\langle y, x \rangle}{\|x\|^2} x, y - \frac{\langle y, x \rangle}{\|x\|^2} x \right\rangle$$

$$\|z\|^2 = \left\langle y, y - \frac{\langle y, x \rangle}{\|x\|^2} x \right\rangle - \left\langle \frac{\langle y, x \rangle}{\|x\|^2} x, y - \frac{\langle y, x \rangle}{\|x\|^2} x \right\rangle$$

(by Axiom (a) of I.P)

$$= \left\langle y, y - \frac{\langle y, x \rangle}{\|x\|^2} x \right\rangle - \frac{\langle y, x \rangle}{\|x\|^2} \left\langle x, y - \frac{\langle y, x \rangle}{\|x\|^2} x \right\rangle$$

(by Axiom (b) of I.P)

$$= \langle y, y \rangle - \left\langle y, \frac{\langle y, x \rangle}{\|x\|^2} x \right\rangle - \left(\frac{\langle y, x \rangle}{\|x\|^2} \right) \left[\langle x, y \rangle - \frac{\langle y, x \rangle}{\|x\|^2} \langle x, x \rangle \right]$$

(by Property of I.P)

$$= \|y\|^2 - \frac{\overline{\langle y, x \rangle}}{\|x\|^2} \langle y, x \rangle - \frac{\langle y, x \rangle}{\|x\|^2} \overline{\langle y, x \rangle} + \left[\frac{\langle y, x \rangle}{\|x\|^2} \frac{\overline{\langle y, x \rangle}}{\|x\|^2} \langle x, x \rangle \right]$$

$$\|z\|^2 = \|y\|^2 - \frac{|\langle y, x \rangle|^2}{\|x\|^2} - \frac{|\langle y, x \rangle|^2}{\|x\|^2} + \frac{|\langle y, x \rangle|^2}{\|x\|^2} \frac{1}{\|x\|^2} \|x\|^2$$

Since $\|z\| \geq 0$.

$$\|z\|^2 = \|z\|^2$$

$$\|y\|^2 - \frac{|\langle y, x \rangle|^2}{\|x\|^2} \geq 0$$

$$\|y\|^2 \geq \frac{|\langle y, x \rangle|^2}{\|x\|^2}$$

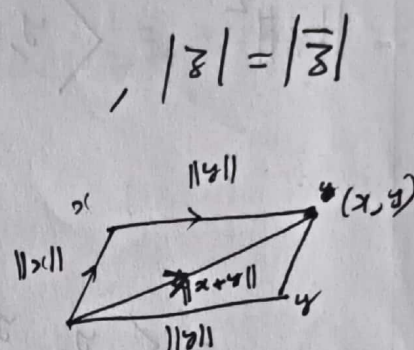
$$\Rightarrow |\langle y, x \rangle|^2 \leq \|x\|^2 \|y\|^2$$

$$\Rightarrow |\langle x, y \rangle|^2 \leq \|x\|^2 \|y\|^2$$

Squaring on both side

$$\Rightarrow |\langle x, y \rangle| \leq \|x\| \|y\|$$

Hence Proved



② Triangle Inequality

If x, y are vectors in an inner product space

then $\|x+y\| \leq \|x\| + \|y\|$

Proof:

Consider

$$\|x+y\|^2 = \langle x+y, x+y \rangle$$

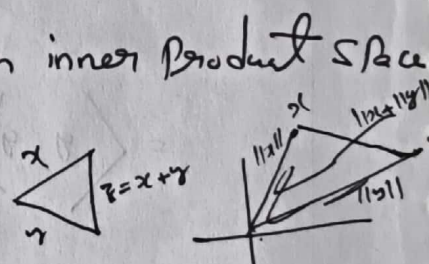
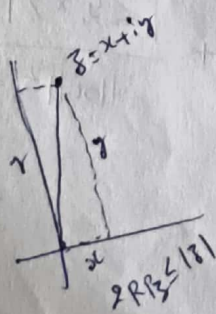
$$= \langle x, x+y \rangle + \langle y, x+y \rangle$$

$$= \langle x, x \rangle + \langle x, y \rangle + \langle y, x \rangle + \langle y, y \rangle$$

$$= \|x\|^2 + \langle x, y \rangle + \overline{\langle x, y \rangle} + \|y\|^2$$

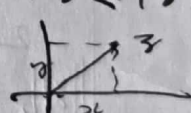
$$= \|x\|^2 + 2 \operatorname{R.P}\{\langle x, y \rangle\} + \|y\|^2$$

$$\leq \|x\|^2 + 2 |\langle x, y \rangle| + \|y\|^2$$



Proposition 4. Cauchy-Schwarz inequality

$$\begin{aligned} \bar{z} &= x+iy, \quad \bar{\bar{z}} = x-iy \\ \bar{z} + \bar{\bar{z}} &= 2x \\ 2\bar{z} + \bar{\bar{z}} &= 2[\operatorname{R.P}\{z\}] \\ \bar{z} &\leq \operatorname{R.P}\{z\} \leq |z| \end{aligned}$$



$$\leq \|x\|^2 + 2\|x\|\|y\| + \|y\|^2 \quad (\text{by Cauchy Schwarz Inequality})$$

$$= (\|x\| + \|y\|)^2$$

$$\Rightarrow \|x+y\|^2 \leq (\|x\| + \|y\|)^2$$

⇒ Taking square root on both side

we get

$$\|x+y\| \leq \|x\| + \|y\|$$

Hence Proved

③ Polar Identities:-

Let V be an inner product space over F . Prove the

For all $x, y \in V$.

$$(a) \langle x, y \rangle = \frac{1}{4} \left[\|x+y\|^2 - \|x-y\|^2 \right] \quad \text{if } F = \mathbb{R}$$

$$(b) \langle x, y \rangle = \frac{1}{4} \sum_{k=1}^4 i^k \|x + i^k y\|^2$$

$$= \frac{1}{4} \left[i \|x + iy\|^2 - \|x - y\|^2 - i \|x - iy\|^2 + \|x + y\|^2 \right]$$

$$= \frac{1}{4} \left[(\|x+y\|^2 - \|x-y\|^2) + i (\|x+iy\|^2 - \|x-iy\|^2) \right] \quad \begin{matrix} F = \mathbb{C} \\ (i^2 = -1) \end{matrix}$$

Proof:

$$(a) \text{ Given that } F = \mathbb{R}, \Rightarrow \langle x, y \rangle = \overline{\langle y, x \rangle} = \langle y, x \rangle \quad \text{--- ①}$$

Consider

$$\begin{aligned}\|x+y\|^2 &= \langle x+y, x+y \rangle \\&= \langle x, x+y \rangle + \langle y, x+y \rangle \\&= \langle x, x \rangle + \langle x, y \rangle + \langle y, x \rangle + \langle y, y \rangle \\&= \|x\|^2 + \langle x, y \rangle + \langle x, y \rangle + \|y\|^2 \quad (\because \text{by } \textcircled{1}) \\&= \|x\|^2 + 2\langle x, y \rangle + \|y\|^2 \quad \text{--- } \textcircled{2}\end{aligned}$$

Also

$$\begin{aligned}\|x-y\|^2 &= \langle x-y, x-y \rangle \\&= \langle x, x-y \rangle - \langle y, x-y \rangle \\&= \langle x, x \rangle - \langle x, y \rangle - \{ \langle y, x \rangle - \langle y, y \rangle \} \\&= \|x\|^2 - \langle x, y \rangle - \langle x, y \rangle + \|y\|^2 \quad (\because \text{by } \textcircled{1}) \\&= \|x\|^2 - 2\langle x, y \rangle + \|y\|^2 \quad \text{--- } \textcircled{3}\end{aligned}$$

$$\textcircled{2} - \textcircled{3} \Rightarrow$$

$$\|x+y\|^2 - \|x-y\|^2 = 4\langle x, y \rangle$$

$$\Rightarrow \langle x, y \rangle = \frac{1}{4} \left[\|x+y\|^2 - \|x-y\|^2 \right]$$

Hence proved

(b) we need to prove

$$\|x+y\|^2 - \|x-y\|^2 + i\|x+iy\|^2 - i\|x-iy\|^2 = 4\langle x, y \rangle$$

$$+ i\left[\|x+iy\|^2 - \|x-iy\|^2\right]$$

Consider,

$$\|x-y\|^2 = \langle x-y, x-y \rangle$$

$$\frac{1}{z} + \overline{\frac{1}{z}} = 2\operatorname{Re} z$$

$$= \langle x, x-y \rangle - \langle y, x-y \rangle$$

$$= \langle x, x \rangle - \langle x, y \rangle - \{ \langle y, x \rangle - \langle y, y \rangle \}$$

$$= \|x\|^2 - \langle x, y \rangle - \overline{\langle x, y \rangle} + \|y\|^2$$

$$= \|x\|^2 - \{ \langle x, y \rangle + \overline{\langle x, y \rangle} \} + \|y\|^2$$

$$= \|x\|^2 - 2 \cdot \operatorname{Re} \{ \langle x, y \rangle \} + \|y\|^2 \quad \text{--- (1)}$$

By

$$\|x+y\|^2 = \|x\|^2 + 2 \cdot \operatorname{Re} \{ \langle x, y \rangle \} + \|y\|^2 \quad \text{--- (2)}$$

(2) - (1) $\Rightarrow$

$$\|x+y\|^2 - \|x-y\|^2 = 4 \cdot \operatorname{Re} \{ \langle x, y \rangle \} \quad \text{--- (3)}$$

Now consider

$$\|x+iy\|^2 = \langle x+iy, x+iy \rangle$$

$$= \langle x, x+iy \rangle + \langle iy, x+iy \rangle$$

$$= \langle x, x \rangle + \langle x, iy \rangle + \langle iy, x \rangle + \langle iy, iy \rangle$$

$$= \|x\|^2 + \overline{i} \langle x, y \rangle + i \overline{\langle x, y \rangle} + i \cdot \overline{i} \langle y, y \rangle$$



$$\begin{aligned}
 \|x+iy\|^2 &= \|x\|^2 - i\langle x, y \rangle + i\overline{\langle x, y \rangle} - (-1)\|y\|^2 \\
 &= \|x\|^2 - i\left[\langle x, y \rangle - \overline{\langle x, y \rangle}\right] + \|y\|^2 \left[\overline{z} - z = 2i \operatorname{Im} z \right] \\
 &= \|x\|^2 - i\left[2i \operatorname{Im}\{\langle x, y \rangle\} \right] + \|y\|^2
 \end{aligned}$$

$$\|x+iy\|^2 = \|x\|^2 + 2 \operatorname{Im}\{\langle x, y \rangle\} + \|y\|^2$$

Also
And

$$i\|x+iy\|^2 = i\|x\|^2 + 2i \operatorname{Im}\{\langle x, y \rangle\} + i\|y\|^2 \quad \text{--- (4)}$$

Also

$$\begin{aligned}
 \|x-iy\|^2 &= \langle x-iy, x-iy \rangle \\
 &= \langle x, x-iy \rangle - \langle iy, x-iy \rangle \\
 &= \langle x, x \rangle - \langle x, iy \rangle - \left\{ \langle iy, x \rangle - \langle iy, iy \rangle \right\} \\
 &= \|x\|^2 - \overline{i} \langle x, y \rangle - i \langle y, x \rangle + i \cdot \overline{i} \langle y, y \rangle \\
 &= \|x\|^2 + i \langle x, y \rangle - i \overline{\langle x, y \rangle} - (-1)\|y\|^2 \\
 &= \|x\|^2 + i\left[2i \operatorname{Im}\{\langle x, y \rangle\} \right] + \|y\|^2 \\
 &= \|x\|^2 - 2 \operatorname{Im}\{\langle x, y \rangle\} + \|y\|^2
 \end{aligned}$$

Ans

$$i\|x-iy\|^2 = i\|x\|^2 - 2i\operatorname{Im}\{\langle x, y \rangle\} + i\|y\|^2 \quad (5)$$

(4) - (5) $\Rightarrow$

$$i\|x+iy\|^2 - i\|x-iy\|^2 = 4i\operatorname{Im}\{\langle x, y \rangle\} \quad (6)$$

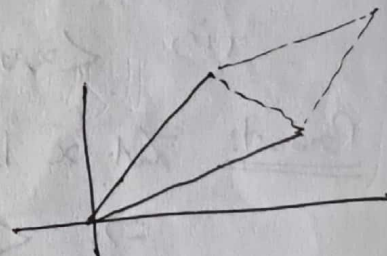
Now

(5) + (6) $\Rightarrow$

$$\begin{aligned} \|x+y\|^2 - \|x-y\|^2 + i\|x+iy\|^2 - i\|x-iy\|^2 \\ = 4\{ \operatorname{Re}\{\langle x, y \rangle\} + i\operatorname{Im}\{\langle x, y \rangle\} \} \\ = 4\langle x, y \rangle \end{aligned}$$

$$\begin{cases} z = x+iy \\ z = \operatorname{Re} z + i\operatorname{Im} z \end{cases}$$

Hence proved

~~Some Important Equalities:-~~(4) Parallelogram law:-In an Inner Product space V , we have

$$\|x+y\|^2 + \|x-y\|^2 = 2\|x\|^2 + 2\|y\|^2 \quad \forall x, y \in V$$

Proof:

From the Previous Problem, w.k.T

$$\|x+y\|^2 = \|x\|^2 + 2\operatorname{Re}\{\langle x, y \rangle\} + \|y\|^2$$

$$\|x-y\|^2 = \|x\|^2 - 2\operatorname{Re}\{\langle x, y \rangle\} + \|y\|^2$$

Then we have

$$\|x+y\|^2 + \|x-y\|^2 = 2\|x\|^2 + 2\|y\|^2;$$

② Pythagorean Theorem :-

~~Let V be~~

Pythagorean Theorem

Refer internet page of
where I.P.S Topic starts.

Pythagorean Theorem:

Let V be a Real inner product space & $x, y \in V$
Then x & y are orthogonal iff $\|x+y\|^2 = \|x\|^2 + \|y\|^2$

Part I:

Given that $\langle x, y \rangle = 0$ — (1)

$$\begin{aligned}\|x+y\|^2 &= \|x\|^2 + 2\operatorname{Re}\langle x, y \rangle + \|y\|^2 \\ &= \|x\|^2 + 2\langle x, y \rangle + \|y\|^2 \\ &= \|x\|^2 + \|y\|^2 \text{ (by (1))}\end{aligned}$$

Part II

Given that $\|x+y\|^2 = \|x\|^2 + \|y\|^2$ — (2)

$$\begin{aligned}\text{Now } \|x+y\|^2 &= \|x\|^2 + 2\langle x, y \rangle + \|y\|^2 \\ \|x\|^2 + \|y\|^2 &= \|x\|^2 + 2\langle x, y \rangle + \|y\|^2 \text{ (by (2))}\end{aligned}$$

$$0 = 2\langle x, y \rangle$$

$$\Rightarrow \langle x, y \rangle = 0$$

$\Rightarrow x$ & y are orthogonal. $\checkmark$

Chapter 2 - Gram Schmidt Orthogonalization Process

Defn Orthogonality:-

Let x & y be vectors in an inner product space V .
Then x is said to be orthogonal to y if,

$$\langle x, y \rangle = 0$$

Result 1 Relation of orthogonality is Symmetric
(ie) $\langle x, y \rangle = 0 \Rightarrow \langle y, x \rangle = 0$

Proof: Let x be orthogonal to y .

$$\Rightarrow \langle x, y \rangle = 0$$

$$\Rightarrow \overline{\langle x, y \rangle} = \overline{0}$$

$$\Rightarrow \langle y, x \rangle = 0$$

$$\Rightarrow y \text{ is orthogonal to } x$$

Result 2 If x is orthogonal to y , then every scalar multiple of x is orthogonal to y .

Proof: Let x be orthogonal to $y \Rightarrow \langle x, y \rangle = 0$

$$\begin{aligned} \text{Now } \langle kx, y \rangle &= k \langle x, y \rangle \\ &= k \cdot 0 \\ &= 0 \end{aligned}$$

$$\Rightarrow kx \text{ is orthogonal to } y.$$

Result 3.1.

Property
The zero vector is the only vector which is orthogonal to itself.

Proof: Assume x is orthogonal to x

$$\Rightarrow \langle x, x \rangle = 0$$

$$\Rightarrow \boxed{x = 0}$$

_____ x _____

Defn orthogonal set:-

A subset S of an inner product space V is said to be orthogonal if any two distinct vectors in S are orthogonal

_____ x _____

Example

Consider the set $u = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$, $v = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$, $w = \begin{bmatrix} -5 \\ -2 \\ 1 \end{bmatrix}$

Suppose $\langle u, v \rangle = uv^T$

$$\text{Here } \langle u, v \rangle = 1(0) + (-2)(1) + (1)(2) = 0 - 2 + 2 = 0$$

$$\langle u, w \rangle = 1(-5) + (-2)(-2) + (1)(1) = -5 + 4 + 1 = 0$$

$$\langle v, w \rangle = 0(-5) + (1)(-2) + 2(1) = 0 - 2 + 2 = 0$$

Here $\{u, v, w\}$ is an orthogonal set.

Example 1:-

Consider $u = \begin{bmatrix} 5 \\ -4 \\ 0 \\ 3 \end{bmatrix}$, $v = \begin{bmatrix} -4 \\ 1 \\ -3 \\ 8 \end{bmatrix}$, $w = \begin{bmatrix} 3 \\ 3 \\ 5 \\ -1 \end{bmatrix}$

$$\langle v, w \rangle = (-4)(3) + (1)(3) + (-3)(5) + (8)(-1)$$

$$= -12 + 3 - 15 - 8$$

$$= -32$$

$$\neq 0$$

$$5(-4) + (-4) + 0 + 29 \\ -20 - 4 + 29 = 0$$

$$15 - 12 + 0 - 3 \\ 3 - 3 = 0$$

Thus $\{u, v, w\}$ ~~are~~ is not an orthogonal set.

— x —

Def: Orthonormal set

Let S be a set of vectors in an inner product space V . Then S is said to be a orthonormal set,

$$\text{if (i) } x \in S \Rightarrow \|x\| = 1 \Rightarrow \sqrt{\langle x, x \rangle} = 1 \\ \Rightarrow \langle x, x \rangle = 1$$

$$\text{(ii) } x, y \in S \text{ \& } x \neq y \Rightarrow \langle x, y \rangle = 0$$

Thus an orthonormal set is an orthogonal set with the additional property that each vector is of unit length.

— x —

Kronecker Delta:

A finite set $S = \{v_1, v_2, \dots, v_n\}$ is orthonormal

$$\text{if } \langle v_i, v_j \rangle = \delta_{ij} = \begin{cases} 1 & \text{if } i=j \\ 0 & \text{if } i \neq j \end{cases}$$

is known as Kronecker delta.

Note 1: An orthonormal set cannot contain a zero vector as $\|0\| = 0$. ($\|x\| = 0$ iff $x = 0$)

Note 2: Every inner product space V which is not equal to zero space possesses an orthonormal set.

Proof:

Algorithm for Gram-Schmidt orthonormalisation Process.

Given Basis $\beta = \{u_1, u_2, u_3, \dots, u_n\}$ &

orthonormal Basis that we are going to find using G.S.O Process.
is represents as $\beta_{ON} = \{v_1, v_2, \dots, v_n\}$

Step 1 $v_1 = \frac{u_1}{\|u_1\|}$

Step 2 $w_2 = u_2 - \langle u_2, v_1 \rangle v_1$
& $v_2 = \frac{w_2}{\|w_2\|}$

Step 3 $w_3 = u_3 - \langle u_3, v_1 \rangle v_1 - \langle u_3, v_2 \rangle v_2$
 $v_3 = \frac{w_3}{\|w_3\|}$

Step 4 $w_4 = u_4 - \langle u_4, v_1 \rangle v_1 - \langle u_4, v_2 \rangle v_2 - \langle u_4, v_3 \rangle v_3$
 $v_4 = \frac{w_4}{\|w_4\|}$

& so on - - - - -

Corollary

Let V be a non-zero finite dimensional inner product space with basis $\{u_1, u_2, \dots, u_n\}$. Then V has an orthonormal basis $\beta = \{v_1, v_2, \dots, v_n\}$.

Furthermore, if $\beta = \{v_1, v_2, \dots, v_n\}$ is an orthonormal basis then $x \in V$ can be expressed as

$$x = \sum_{i=1}^n \langle x, v_i \rangle v_i$$

where $\langle x, v_i \rangle$ is called Fourier coefficient of x relative to β . $v_1 = (0, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$ $\|v_1\| = \sqrt{2}$ (2 marks)

$v_2 = (1, -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$ $\langle v_2, v_1 \rangle = \frac{1}{\sqrt{2}}$
 $\|v_2\| = \sqrt{3}/\sqrt{2}$ $v_2 = (\frac{2}{\sqrt{6}}, -\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}})$

Problem 1

Apply Gram Schmidt process to the vector given below to obtain an orthonormal basis β for V &

Compute the Fourier coefficient of the given vector x relative to β . $\langle v_3, v_1 \rangle = \frac{1}{\sqrt{2}}$ $\langle v_3, v_2 \rangle = \frac{1}{\sqrt{6}}$ $v_3 = (\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}})$ $\|v_3\| = \frac{2}{\sqrt{3}}$ (6 marks)

(i) $V = \mathbb{R}^3$, $S = \{(1, 1, 1), (0, 1, 1), (0, 0, 1)\}$ and $x = (1, 0, 1)$

$S = \{(1, 0, 1), (0, 1, 1), (1, 3, 3)\}$ & $x = (1, 1, 2)$

(ii) $V = P_2(\mathbb{R})$ with Inner Product

$\langle f(x), g(x) \rangle = \int_0^1 f(x)g(x)dx$ $\langle x, v_1 \rangle = 3/\sqrt{2}$ $\langle x, v_2 \rangle = \frac{3}{\sqrt{6}}$ $\langle x, v_3 \rangle = 0$

$S = \{1, x, x^2\}$ & $h(x) = 1+x$

(iii) $V = M_{2 \times 2}(\mathbb{R})$, $S = \left\{ \begin{pmatrix} 3 & 5 \\ -1 & 1 \end{pmatrix}, \begin{pmatrix} -1 & 9 \\ 5 & -1 \end{pmatrix}, \begin{pmatrix} 7 & -17 \\ 2 & -6 \end{pmatrix} \right\}$

& $A = \begin{bmatrix} -1 & 2 & 7 \\ -4 & 8 \end{bmatrix}$

(i) Given that $V = \mathbb{R}^3$

Let $x = (a_1, a_2, a_3)$ & $y = (b_1, b_2, b_3)$, $a_i, b_i \in \mathbb{R}$

Inner Product is defined $\mathbb{R}^3$ is

$$\langle x, y \rangle = a_1 b_1 + a_2 b_2 + a_3 b_3$$

Here $S = \{ (1, 1, 1), (0, 1, 1), (0, 0, 1) \}$

(i.e. $u_1 = (1, 1, 1)$, $u_2 = (0, 1, 1)$, $u_3 = (0, 0, 1)$)

Step 1 :-

$$v_1 = \frac{u_1}{\|u_1\|}, \text{ where } \|u_1\|^2 = \langle u_1, u_1 \rangle$$

$$= 1^2 + 1^2 + 1^2$$

$$= 3$$

$$\|u_1\| = \sqrt{3}$$

$$v_1 = \frac{1}{\sqrt{3}} (1, 1, 1)$$

$$= \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right) \text{ --- (1)}$$

Step 2

$$w_2 = u_2 - \langle u_2, v_1 \rangle v_1$$

Now $\langle u_2, v_1 \rangle = (0, 1, 1) \cdot \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right)$

$$= 0 + \frac{1}{\sqrt{3}} + \frac{1}{\sqrt{3}}$$

$$= \frac{2}{\sqrt{3}}$$

Now

$$w_2 = (0, 1, 1) - \frac{2}{\sqrt{3}} \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right)$$

$$w_2 = (0, 1, 1) - \left(\frac{2}{3}, \frac{2}{3}, \frac{2}{3}\right)$$

$$= \left(-\frac{2}{3}, \frac{1}{3}, \frac{1}{3}\right)$$

Now $v_2 = \frac{w_2}{\|w_2\|}$, where $\|w_2\|^2 = \langle w_2, w_2 \rangle$

$$= \left(-\frac{2}{3}\right)^2 + \left(\frac{1}{3}\right)^2 + \left(\frac{1}{3}\right)^2$$

$$= \frac{4}{9} + \frac{1}{9} + \frac{1}{9}$$

$$= \frac{6}{9} = \frac{2}{3}$$

$$\|w_2\| = \sqrt{\frac{2}{3}}$$

Now $v_2 = \frac{\sqrt{3}}{\sqrt{2}} \left(-\frac{2}{3}, \frac{1}{3}, \frac{1}{3}\right)$

$$= \frac{\sqrt{3}}{\sqrt{2} \cdot 3} (-2, 1, 1)$$

$$= \frac{1}{\sqrt{6}} (-2, 1, 1)$$

$$v_2 = \left(-\frac{2}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}\right) \text{ --- (2)}$$

Step 3 :-

$$w_3 = u_3 - \langle u_3, v_1 \rangle v_1 - \langle u_3, v_2 \rangle v_2$$

Now $\langle u_3, v_1 \rangle = (0, 0, 1) \cdot \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$

$$= 0 + 0 + \frac{1}{\sqrt{3}}$$

$$= \frac{1}{\sqrt{3}}$$

$$\begin{aligned}\langle u_3, v_2 \rangle &= (0, 0, 1) \cdot \left(\frac{-2}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}} \right) \\ &= 0 + 0 + \frac{1}{\sqrt{6}} \\ &= \frac{1}{\sqrt{6}}\end{aligned}$$

$$\begin{aligned}w_3 &= (0, 0, 1) - \frac{1}{\sqrt{3}} \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right) - \frac{1}{\sqrt{6}} \left(\frac{-2}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}} \right) \\ &= (0, 0, 1) - \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3} \right) - \left(\frac{-2}{6}, \frac{1}{6}, \frac{1}{6} \right) \\ &= \left(0 - \frac{1}{3} + \frac{2}{6}, 0 - \frac{1}{3} - \frac{1}{6}, 1 - \frac{1}{3} - \frac{1}{6} \right) \\ &= \left(0, -\frac{1}{2}, \frac{1}{2} \right)\end{aligned}$$

$$v_3 = \frac{w_3}{\|w_3\|}, \quad \text{where } \|w_3\|^2 = \langle w_3, w_3 \rangle$$

$$= 0 + \left(-\frac{1}{2} \right)^2 + \left(\frac{1}{2} \right)^2$$

$$= \frac{1}{2}$$

$$= \frac{1}{\sqrt{2}} \left(0, -\frac{1}{2}, \frac{1}{2} \right) \quad \|w_3\| = \frac{1}{\sqrt{2}}$$

$$= \sqrt{2} \left(0, -\frac{1}{2}, \frac{1}{2} \right)$$

$$= \left(0, -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right) \quad \text{--- (3)}$$

From (1) (2) (3), we have found an orthonormal basis

$$\beta = \{v_1, v_2, v_3\} = \left\{ \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right), \left(\frac{-2}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}} \right), \left(0, -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right) \right\}$$

To find Fourier Coeff of $x = (1, 0, 1)$ relative to β

We have

$$\begin{aligned}\langle x, v_1 \rangle &= (1, 0, 1) \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right) \\ &= \frac{1}{\sqrt{3}} + 0 + \frac{1}{\sqrt{3}} = \frac{2}{\sqrt{3}}\end{aligned}$$

$$\begin{aligned}\langle x, v_2 \rangle &= (1, 0, 1) \left(\frac{-2}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}} \right) \\ &= \frac{-2}{\sqrt{6}} + \frac{1}{\sqrt{6}} = \frac{-1}{\sqrt{6}}\end{aligned}$$

$$\begin{aligned}\langle x, v_3 \rangle &= (1, 0, 1) \left(0, -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right) \\ &= 0 + 0 + \frac{1}{\sqrt{2}} \\ &= \frac{1}{\sqrt{2}}\end{aligned}$$

Thus $\frac{2}{\sqrt{3}}, \frac{-1}{\sqrt{6}}, \frac{1}{\sqrt{2}}$ are Fourier coeff of $x = (1, 0, 1)$ relative to β .

Verification...

$$x = \langle x, v_1 \rangle v_1 + \langle x, v_2 \rangle v_2 + \langle x, v_3 \rangle v_3$$

$$= \frac{2}{\sqrt{3}} \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right) + \left(\frac{-1}{\sqrt{6}} \right) \left(\frac{-2}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}} \right) + \frac{1}{\sqrt{2}} \left(0, -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right)$$

$$= \left(\frac{2}{3}, \frac{2}{3}, \frac{2}{3} \right) + \left(\frac{2}{6}, -\frac{1}{6}, -\frac{1}{6} \right) + \left(0, -\frac{1}{2}, \frac{1}{2} \right)$$

$$= \left(\frac{2}{3} + \frac{1}{3}, \frac{2}{3} - \frac{1}{6} - \frac{1}{2}, \frac{2}{3} - \frac{1}{6} + \frac{1}{2} \right)$$

$$= (1, 0, 1)$$

Have Verified

_____x_____

(ii) Given that $V = P_2(\mathbb{R})$

Let $f(x), g(x) \in P_2(\mathbb{R})$, Then the Inner Product is

$$\langle f(x), g(x) \rangle = \int_0^1 f(x)g(x)dx.$$

Here $S = \{1, x, x^2\}$

$$u_1 = 1, u_2 = x, u_3 = x^2$$

Step 1

$$v_1 = \frac{u_1}{\|u_1\|} = \frac{1}{1} = 1 \quad \text{--- (1)} \quad \begin{aligned} \|u_1\|^2 &= \langle u_1, u_1 \rangle \\ &= \langle 1, 1 \rangle \\ &= \int_0^1 1 \cdot 1 dx = (x)_0^1 \\ &= 1 \end{aligned}$$

Step 2 :-

$$w_2 = u_2 - \langle u_2, v_1 \rangle v_1$$

$$\text{Now } \langle u_2, v_1 \rangle = \langle x, 1 \rangle$$

$$= \int_0^1 (x \cdot 1) dx = \left(\frac{x^2}{2} \right)_0^1 = \frac{1}{2}$$

$$w_2 = x - \frac{1}{2}(1)$$

$$w_2 = x - \frac{1}{2}$$

$$v_2 = \frac{w_2}{\|w_2\|}, \text{ where } \|w_2\|^2 = \langle w_2, w_2 \rangle$$

$$= \int_0^1 (x - \frac{1}{2})(x - \frac{1}{2}) dx$$

$$= \int_0^1 (x - \frac{1}{2})^2 dx$$

$$= \left[\frac{(x - \frac{1}{2})^3}{1.3} \right]_0^1$$

$$= \frac{(\frac{1}{2})^3}{3} - \frac{(-\frac{1}{2})^3}{3}$$

$$= \frac{1}{24} + \frac{1}{24} = \frac{1}{12}$$

$$\therefore v_2 = \frac{1}{(1/\sqrt{2})} (x - 1/2), \text{ Since } \|u_2\| = \frac{1}{\sqrt{2}} = \frac{1}{2\sqrt{3}}$$

$$= 2\sqrt{3} (x - 1/2) \quad \text{--- (2)}$$

Step 3:

$$u_3 = u_3 - \langle u_3, v_2 \rangle v_2 - \langle u_3, v_1 \rangle v_1$$

For,

$$\langle u_3, v_2 \rangle = \langle x^2, 2\sqrt{3}(x - 1/2) \rangle$$

$$= \int_0^1 x^2 \cdot 2\sqrt{3}(x - 1/2) dx$$

$$= 2\sqrt{3} \int_0^1 (x^3 - \frac{1}{2}x^2) dx$$

$$= 2\sqrt{3} \left[\frac{x^4}{4} - \frac{1}{2} \frac{x^3}{3} \right]_0^1$$

$$= 2\sqrt{3} \left[\left(\frac{1}{4} - \frac{1}{6} \right) - (0) \right]$$

$$= 2\sqrt{3} \left[\frac{6-4}{24} \right]$$

$$= 2\sqrt{3} \left[\frac{2}{24} \right]$$

$$= \frac{\sqrt{3}}{6}$$

$$= \frac{\sqrt{3}}{2 \times 3}$$

$$= \frac{1}{2\sqrt{3}}$$

$$\therefore \langle u_3, v_1 \rangle = \langle x^2, 1 \rangle$$

$$= \int_0^1 x^2 \cdot 1 dx = \left(\frac{x^3}{3} \right)_0^1 = \left(\frac{1}{3} \right)$$

Now

$$\omega_3 = x^2 - \frac{1}{2\sqrt{3}} (x - \frac{1}{2}) - \frac{1}{3} (1)$$

$$= x^2 - (x - \frac{1}{2}) - \frac{1}{3}$$

$$= x^2 - x + \frac{1}{2} - \frac{1}{3}$$

$$\omega_3 = x^2 - x + \frac{1}{6}$$

$$v_3 = \frac{\omega_3}{\|\omega_3\|}, \text{ For } \|\omega_3\|^2 = \langle \omega_3, \omega_3 \rangle$$

$$= \int_0^1 (x^2 - x + \frac{1}{6})(x^2 - x + \frac{1}{6}) dx$$

$$= \int_0^1 (x^2 - x + \frac{1}{6})^2 dx$$

$$(x^4 - 2x^3 + \frac{4}{3}x^2 - \frac{1}{3}x + \frac{1}{36})$$

$$= \int_0^1 [x^4 + x^2 + \frac{1}{36} - 2x^3 + \frac{1}{3}x - \frac{1}{3}x] dx$$

$$= \frac{x^5}{5} - \frac{2x^4}{4} + \frac{4}{3} \frac{x^3}{3} - \frac{1}{3} \frac{x^2}{2} + \frac{10}{36}x$$

$$= \frac{1}{5} - \frac{1}{2} + \frac{4}{9} - \frac{1}{6} + \frac{1}{36}$$

$$= \frac{1}{5} - \frac{1}{2} + \frac{4}{9} - \frac{1}{6} + \frac{1}{36}$$

$$= \frac{1}{5} + \left(\frac{12 + 1 - 18 + 4 - 6(36)}{36} \right)$$

$$= \frac{1}{5} + \left(\frac{-18 + 16 - 6 + 1}{36} \right)$$

$$= \frac{1}{5} + \left(\frac{17 - 24}{36} \right)$$

$$= \frac{1}{5} + \left(\frac{-7}{36} \right)$$

$$= \frac{1}{5} + \left(\frac{-7}{36} \right)$$

$$= \frac{1}{5} + \frac{7}{36}$$

$$= \frac{36 - 35}{5 \times 36}$$

$$\|\omega_3\|^2 = \frac{1}{5 \times 36}$$

$$\Rightarrow \|\omega_3\| = \frac{1}{6\sqrt{5}}$$

$$S = x^2 - x + \frac{1}{6}$$

$$\frac{dS}{dx} = 2x - 1$$

$$\frac{1}{5}$$

$$v_3 = 6\sqrt{5}(x^2 - x + 1/6) \text{ --- (3)}$$

From (1), (2), (3),

We have found an orthonormal basis

$$\beta = \{v_1, v_2, v_3\} = \left\{ 1, 2\sqrt{3}(x - 1/2), 6\sqrt{5}(x^2 - x + 1/6) \right\}$$

To find Fourier coeff

For $h(x) = 1+x$ relative to β

$$\langle h(x), v_1 \rangle = \int_0^1 (1+t) \cdot 1 \, dt = \left(t + \frac{t^2}{2} \right)_0^1 = 1 + \frac{1}{2} = \frac{3}{2}$$

$$\begin{aligned} \langle h(x), v_2 \rangle &= \int_0^1 (1+t) 2\sqrt{3}(t - 1/2) \, dt \\ &= 2\sqrt{3} \int_0^1 \left(t - 1/2 + t^2 - t/2 \right) \, dt \end{aligned}$$

$$= 2\sqrt{3} \int_0^1 \left(t^2 + \frac{t}{2} - \frac{1}{2} \right) \, dt$$

$$= 2\sqrt{3} \left[\frac{t^3}{3} + \frac{t^2}{4} - \frac{1}{2}t \right]_0^1$$

$$= 2\sqrt{3} \left[\frac{1}{3} + \frac{1}{4} - \frac{1}{2} \right]$$

$$= 2\sqrt{3} \left[\frac{4+3-6}{12} \right]$$

$$= \frac{\sqrt{3}}{6} = \frac{\sqrt{3}}{2 \times 3} = \frac{1}{2\sqrt{3}} //$$

$$\begin{aligned}
 \langle h(x), v_3 \rangle &= \int_0^1 (1+t) 6\sqrt{2} (t^2 - t + \frac{1}{6}) dt \quad \frac{1}{6} - 1 = -\frac{5}{6} \\
 &= 6\sqrt{2} \int_0^1 (t^2 - t + \frac{1}{6} + t^3 - t^2 + \frac{t}{6}) dt \\
 &= 6\sqrt{2} \int_0^1 (t^3 - \frac{5}{6}t + \frac{1}{6}) dt \\
 &= 6\sqrt{2} \left[\frac{t^4}{4} - \frac{5}{6} \frac{t^2}{2} + \frac{1}{6} t \right]_0^1 \\
 &= 6\sqrt{2} \left[\left(\frac{1}{4} - \frac{5}{12} + \frac{1}{6} \right) - (0) \right] \\
 &= 6\sqrt{2} \left[\frac{3 - 5 + 2}{12} \right] \\
 &= \frac{\sqrt{2}}{2} [0] \\
 &= 0
 \end{aligned}$$

Thus $\left\{ \frac{3}{2}, \frac{1}{2\sqrt{2}}, 0 \right\}$ are Fourier coeffs of $h(x) = 1+x$ relative to β .

Verification:-

$$\begin{aligned}
 h(x) &= \langle h(x), v_1 \rangle v_1 + \langle h(x), v_2 \rangle v_2 + \langle h(x), v_3 \rangle v_3 \\
 &= \frac{3}{2} \cdot 1 + \frac{1}{2\sqrt{2}} 2\sqrt{2} (x - \frac{1}{2}) + 0 \cdot (6\sqrt{2}) (x^2 - x + \frac{1}{6}) \\
 &= \frac{3}{2} + x - \frac{1}{2} \\
 &= x + 1 //
 \end{aligned}$$

(iii)

Given that

$$V = M_{2 \times 2}(\mathbb{R})$$

Let $A, B \in M_{2 \times 2}(\mathbb{R})$, then theinner product, defined on $M_{2 \times 2}(\mathbb{R})$ is

$$\langle A, B \rangle = \text{tr}(A^* B)$$

$$A^* = (\overline{A})^T$$

$$= \text{tr}(A^T B), \quad (\text{Since Here } F = \mathbb{R})$$

Here

$$u_1 = \begin{pmatrix} 3 & 5 \\ -1 & 1 \end{pmatrix}, \quad u_2 = \begin{pmatrix} -1 & 9 \\ 5 & -1 \end{pmatrix}, \quad u_3 = \begin{pmatrix} 7 & -17 \\ 2 & -6 \end{pmatrix}$$

Step 1:-

$$v_1 = \frac{u_1}{\|u_1\|},$$

$$\text{where } \|u_1\|^2 = \langle u_1, u_1 \rangle$$

$$= \text{tr}(u_1^T u_1)$$

$$\text{For } u_1^T = \begin{pmatrix} 3 & -1 \\ 5 & 1 \end{pmatrix}, \quad u_1 = \begin{pmatrix} 3 & 5 \\ -1 & 1 \end{pmatrix}$$

$$u_1^T u_1 = \begin{pmatrix} 3 & -1 \\ 5 & 1 \end{pmatrix} \begin{pmatrix} 3 & 5 \\ -1 & 1 \end{pmatrix} = \begin{bmatrix} 9+1 & 15-1 \\ 15-1 & 25+1 \end{bmatrix} = \begin{bmatrix} 10 & 14 \\ 14 & 26 \end{bmatrix}$$

$$\therefore \|u_1\|^2 = \text{tr} \begin{bmatrix} 10 & 14 \\ 14 & 26 \end{bmatrix} = 36$$

$$\boxed{\|u_1\| = 6}$$

$$\text{Hence } v_1 = \frac{1}{6} \begin{bmatrix} 3 & 5 \\ -1 & 1 \end{bmatrix} \quad \text{--- (1)}$$

Step 2

$$w_2 = u_2 - \langle u_2, v_1 \rangle v_1$$

$$\text{tr}(AB) = \text{tr}(BA)$$

where $\langle u_2, v_1 \rangle = \text{tr}(u_2^T v_1)$

$$u_2^T v_1 = \frac{1}{6} \begin{bmatrix} -1 & 5 \\ 9 & -1 \end{bmatrix} \begin{bmatrix} -3 & 5 \\ -1 & 1 \end{bmatrix}$$

$$= \frac{1}{6} \begin{bmatrix} -3-5 & -5+5 \\ 27+1 & 45-1 \end{bmatrix}$$

$$= \frac{1}{6} \begin{bmatrix} -8 & 0 \\ 28 & 44 \end{bmatrix}$$

$$\langle u_2, v_1 \rangle = \text{tr} \begin{bmatrix} -8/6 & 0 \\ 28/6 & 44/6 \end{bmatrix}$$

$$= -\frac{8}{6} + \frac{44}{6} = \frac{36}{6} = 6$$

$$w_2 = \begin{bmatrix} -1 & 9 \\ 5 & -1 \end{bmatrix} - 6 \frac{1}{6} \begin{bmatrix} -3 & 5 \\ -1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -4 & 4 \\ 6 & -2 \end{bmatrix}$$

$$v_2 = \frac{w_2}{\|w_2\|}$$

where $\|w_2\|^2 = \langle w_2, w_2 \rangle$

$$= \text{tr} \langle w_2^T w_2 \rangle$$

$$w_2^T w_2 = \begin{bmatrix} -4 & 6 \\ 4 & -2 \end{bmatrix} \begin{bmatrix} -4 & 4 \\ 6 & -2 \end{bmatrix} = \begin{bmatrix} 16+36 & -16-12 \\ -16-12 & 16+4 \end{bmatrix} = \begin{bmatrix} 52 & -28 \\ -28 & 20 \end{bmatrix}$$

$$= \text{tr} \begin{bmatrix} 52 & -28 \\ -28 & 20 \end{bmatrix}$$

$$= 72$$

$$\|u_2\| = \sqrt{72} = \sqrt{36 \times 2} = 6\sqrt{2}$$

$$v_2 = \frac{1}{6\sqrt{2}} \begin{bmatrix} -4 & 4 \\ 6 & -2 \end{bmatrix} \quad \text{--- (2)}$$

Step 3:

$$u_3 = u_3 - \langle u_3, v_1 \rangle v_1 - \langle u_3, v_2 \rangle v_2$$

Consider

$$\langle u_3, v_1 \rangle = \frac{1}{6} (u_3^T v_1)$$

$$u_3^T v_1 = \frac{1}{6} \begin{bmatrix} 7 & 2 \\ -17 & -6 \end{bmatrix} \begin{bmatrix} 3 & 5 \\ -1 & 1 \end{bmatrix}$$

$$= \frac{1}{6} \begin{bmatrix} 21-2 & 35+2 \\ -51+6 & -85-6 \end{bmatrix}$$

$$= \frac{1}{6} \begin{bmatrix} 19 & 37 \\ -45 & -91 \end{bmatrix}$$

$$\text{Now } \langle u_3, v_1 \rangle = \frac{1}{6} \begin{bmatrix} 19 & 37 \\ -45 & -91 \end{bmatrix}$$

$$= \frac{19}{6} - \frac{91}{6}$$

$$= \frac{-72}{6}$$

$$= -12$$

$$\langle u_3, v_2 \rangle = \frac{1}{6\sqrt{2}} (u_3^T v_2)$$

$$u_3^T v_2 = \frac{1}{6\sqrt{2}} \begin{bmatrix} 7 & 2 \\ -17 & -6 \end{bmatrix} \begin{bmatrix} -4 & 4 \\ 6 & -2 \end{bmatrix}$$

$$= \frac{1}{6\sqrt{2}} \begin{bmatrix} -28+12 & 28-4 \\ 68-36 & -68+12 \end{bmatrix} = \frac{1}{6\sqrt{2}} \begin{bmatrix} -16 & 24 \\ 32 & -56 \end{bmatrix}$$

$$\frac{74 \times 2}{68} = \frac{36}{32}$$

$$\frac{27 \times 2}{81}$$

$$\frac{27 \times 9}{27} = \frac{27}{293}$$



$$\langle u_3, v_2 \rangle = \frac{1}{6\sqrt{2}} \begin{bmatrix} -16 & 24 \\ 32 & -56 \end{bmatrix}$$

$$= \frac{-16}{6\sqrt{2}} - \frac{56}{6\sqrt{2}}$$

$$= \frac{-72}{6\sqrt{2}}$$

$$= \frac{-(36 \times 2)}{6\sqrt{2}}$$

$$= -6\sqrt{2}$$

$$\begin{aligned} \text{Now } \omega_3 &= \begin{bmatrix} 7 & -17 \\ 2 & -6 \end{bmatrix} - (-\frac{2}{\sqrt{2}}) \frac{1}{6} \begin{bmatrix} 3 & 5 \\ -1 & 1 \end{bmatrix} - (-6\sqrt{2}) \frac{1}{6\sqrt{2}} \begin{bmatrix} -4 & 4 \\ 6 & -2 \end{bmatrix} \\ &= \begin{bmatrix} 7 & -17 \\ 2 & -6 \end{bmatrix} + \begin{bmatrix} 6 & 10 \\ -2 & 2 \end{bmatrix} + \begin{bmatrix} -4 & 4 \\ 6 & -2 \end{bmatrix} \end{aligned}$$

$$= \begin{bmatrix} 9 & -3 \\ 6 & -6 \end{bmatrix}$$

$$\text{Now } \|\omega_3\|^2 = \langle \omega_3, \omega_3 \rangle = \text{tr}(\omega_3^T \omega_3)$$

$$\begin{aligned} \omega_3^T \omega_3 &= \begin{bmatrix} 9 & 6 \\ -3 & -6 \end{bmatrix} \begin{bmatrix} 9 & -3 \\ 6 & -6 \end{bmatrix} = \begin{bmatrix} 81+36 & -27-36 \\ -27-36 & 9+36 \end{bmatrix} \\ &= \begin{bmatrix} 117 & -63 \\ -63 & 45 \end{bmatrix} \end{aligned}$$

$$\|\omega_3\|^2 = \text{tr} \begin{bmatrix} 117 & -63 \\ -63 & 45 \end{bmatrix} = 162$$

$$\|\omega_3\| = \sqrt{162}$$

$$\begin{array}{r} 1 \\ 117 \\ 45 \\ \hline 162 \end{array}$$

$$\text{Hence } v_3 = \frac{w_3}{\|w_3\|}$$

$$= \frac{1}{\sqrt{162}} \begin{bmatrix} 9 & -3 \\ 6 & -6 \end{bmatrix} \quad \text{--- (3)}$$

Thus from (1), (2), (3) we have constructed,
orthonormal basis

$$\beta = \{v_1, v_2, v_3\} = \left\{ \frac{1}{6} \begin{bmatrix} 3 & 5 \\ -1 & 1 \end{bmatrix}, \frac{1}{6\sqrt{2}} \begin{bmatrix} -4 & 4 \\ 6 & -2 \end{bmatrix}, \frac{1}{\sqrt{162}} \begin{bmatrix} 9 & -3 \\ 6 & -6 \end{bmatrix} \right\}$$

To find Fourier Coeff.

$$\text{Given that } A = \begin{bmatrix} -1 & 27 \\ -4 & 8 \end{bmatrix}$$

$$\frac{2 \times 27 \times 4}{108}$$

$$\langle A, v_1 \rangle = \text{tr}[A^T v_1]$$

$$A^T v_1 = \frac{1}{6} \begin{bmatrix} -1 & -4 \\ 27 & 8 \end{bmatrix} \begin{bmatrix} 3 & 5 \\ -1 & 1 \end{bmatrix} = \frac{1}{6} \begin{bmatrix} 1 & -9 \\ 73 & 143 \end{bmatrix}$$

$$\langle A, v_1 \rangle = \text{tr} \left[\frac{1}{6} \begin{bmatrix} 1 & -9 \\ 73 & 143 \end{bmatrix} \right] = \frac{1}{6} + \frac{143}{6} = \frac{144}{6} = 24$$

$$\langle A, v_2 \rangle = \text{tr}(A^T v_2) = \text{tr} \left[\begin{bmatrix} -1 & -4 \\ 27 & 8 \end{bmatrix} \frac{1}{6\sqrt{2}} \begin{bmatrix} -4 & 4 \\ 6 & -2 \end{bmatrix} \right]$$

$$= \text{tr} \left[\frac{1}{6\sqrt{2}} \begin{bmatrix} 4-24 & -4+8 \\ -108+48 & 108+16 \end{bmatrix} \right]$$

$$= \text{tr} \left[\frac{1}{6\sqrt{2}} \begin{bmatrix} -20 & 4 \\ 60 & 92 \end{bmatrix} \right]$$

$$= \frac{-20}{6\sqrt{2}} + \frac{92}{6\sqrt{2}} = \frac{72}{6\sqrt{2}} = \frac{12}{\sqrt{2}} = \frac{6 \times 2}{\sqrt{2}} = 6$$

$$\langle A, v_3 \rangle = \text{tr} \left[A^T v_3 \right] = \text{tr} \left[\begin{bmatrix} -1 & -4 \\ 27 & 8 \end{bmatrix} \frac{1}{\sqrt{162}} \begin{bmatrix} 9 & -3 \\ 6 & -6 \end{bmatrix} \right]$$

$$= \text{tr} \left[\frac{1}{\sqrt{162}} \begin{bmatrix} -9+24 & 3+24 \\ 243+48 & -81-48 \end{bmatrix} \right]$$

$$= \text{tr} \left[\frac{1}{\sqrt{162}} \begin{bmatrix} -33 & 27 \\ 291 & -129 \end{bmatrix} \right]$$

$$= \frac{-33}{\sqrt{162}} - \frac{129}{\sqrt{162}}$$

$$\frac{129}{33} \\ \frac{162}{162}$$

$$= \frac{-162}{\sqrt{162}}$$

$$= -\sqrt{162}$$

$$= -9\sqrt{2}$$

Thus $\{24, 6\sqrt{2}, -9\sqrt{2}\}$ are Fourier Coeff of A
relative to B

Verification:-

$$A = \langle A, v_1 \rangle v_1 + \langle A, v_2 \rangle v_2 + \langle A, v_3 \rangle v_3$$

$$= 24 \frac{1}{\sqrt{2}} \begin{bmatrix} 3 & 5 \\ -1 & 1 \end{bmatrix} + 6\sqrt{2} \frac{1}{6\sqrt{2}} \begin{bmatrix} -4 & 4 \\ 6 & -2 \end{bmatrix} - 9\sqrt{2} \left(\frac{1}{9\sqrt{2}} \right) \begin{bmatrix} 9 & -3 \\ 6 & -6 \end{bmatrix}$$

$$= \begin{bmatrix} 12 & 20 \\ -4 & 4 \end{bmatrix} + \begin{bmatrix} -4 & 4 \\ 6 & -2 \end{bmatrix} + \begin{bmatrix} -9 & 3 \\ -6 & 6 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & 27 \\ -4 & 8 \end{bmatrix}$$

Have Verified